Replacing beef: nutrition, emissions and the limits of a campus ban

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Universities are removing beef from their menus to cut emissions, but three questions are usually left open. Can the nutrients in beef be replaced, and by what? How large is the climate benefit at the scale of a city? How much of a single institution's ban reaches the atmosphere once diners and markets respond? We answer them for London and for Imperial College London, which removed beef from its in-house catering in October 2024. Beef's protein is easy to replace, but its absorbable zinc and vitamin B_{12} are not: no combination of non-ruminant foods matches them within twice beef's energy without fortified foods or supplements. Nutritionally equivalent baskets emit 72–93% less than the beef they replace (5.59 kg CO_2e per 100 g cooked beef for the UK supply mix). If every Londoner replaced beef this way, consumption-based emissions would fall by 2.8 (2.0–3.7) Mt CO_2e a year. That is equivalent to 10% (7–13%) of London's territorial inventory, although almost all of the saving occurs outside London's boundary. For Imperial, market leakage and off-campus purchasing cut the long-run reduction to 150 t CO_2e a year, 65% of what procurement accounting would report, and to 47 t in the short run. At prices held fixed, the implied welfare cost to diners is about £1,730 (£911–3,725) per tonne avoided, several times the UK government's carbon value. Removing beef cuts emissions when a whole population changes what it eats; a single campus achieves much less than its inventory suggests.

Food systems account for about a third of anthropogenic greenhouse-gas emissions[1]. Ruminant meat is their most emission-intensive major product: per kilogram or per gram of protein, beef from dedicated beef herds emits roughly ten times more than poultry or pork and fifty times more than pulses[2]. Diet change is therefore part of every pathway that keeps warming near 1.5–2 °C[3–5], and UK vegans' dietary footprints are about a quarter of those of high meat-eaters[6]. Institutions have acted on this. The University of Cambridge's catering service removed beef and lamb in 2016 and reported a third less CO_2e per kilogram of food purchased[7]. Goldsmiths, University of London, banned beef in 2019[8]. Imperial College London adopted a policy in 2022 to phase beef out of its catering[9]. Its in-house caterer, Taste Imperial, removed beef from every menu in October 2024, after finding that fresh meat was 8% of catering purchases but 40% of catering emissions[10].

Such policies raise three questions that are rarely answered together. The first is biological: beef is a dense source of high-quality protein, haem iron, zinc and vitamin B_{12}, so can the same nutrition be obtained from other foods, which ones, and in what amounts? The second is about emissions: if beef were replaced by nutritionally equivalent foods, how large would the change be, in percentage terms, in tonnes of CO_2e, and relative to the emissions of a city such as London? The third is economic. A campus is a tiny buyer in a large market, so when it stops buying beef the price falls slightly, other consumers buy more, and some diners buy beef elsewhere. How much of the accounting saving survives, and what does the policy cost the people who eat on campus? We address all three with one open, reproducible model chain (Methods). It links food composition and diet optimisation to life-cycle emissions, a market equilibrium model and a discrete-choice welfare model, and it uses Imperial's own purchase records.

Beef's nutrients can be replaced, but not one-for-one

A 100 g portion of cooked lean beef mince supplies 230 kcal, 28.4 g of protein with a digestible indispensable amino acid score (DIAAS) of 0.99[11], 3.08 mg of iron of which about two-thirds is haem, 6.84 mg of zinc, 2.80 µg of vitamin B_{12} and 21.6 µg of selenium. After adjusting for bioavailability (Methods), these are 28.1 g of quality-adjusted protein, 0.61 mg of absorbable iron, 2.39 mg of absorbable zinc and 1.68 µg of absorbable B_{12}. In the UK, meat and meat products supply about a third of adults' zinc, over a quarter of their vitamin B_{12} and about a fifth of their iron[12]. A quarter of women aged 19–64 already have iron intakes below the lower reference nutrient intake[12].

Protein is the easy part (Fig. 1a). Chicken breast matches beef's quality-adjusted protein with 91 g, tofu with 168 g and mycoprotein with 196 g. Cooked lentils need 537 g because their DIAAS is low[13]. Zinc and vitamin B_{12} are the binding nutrients. Matching beef's absorbable zinc takes 684 g of chicken breast, 270 g of pork loin and 1.0 kg of lentils. The phytate in pulses roughly halves zinc absorption[14]. The exceptions are mycoprotein (137 g), whose zinc content is high and which contains almost no phytate[15,16], and pumpkin seeds (170 g, but over 900 kcal). Vitamin B_{12} occurs naturally only in animal foods. Matching it takes 143 g of salmon, 488 g of milk, 824 g of chicken or 1.7 kg of egg, whose B_{12} is poorly absorbed[17]. It takes 160 g of fortified cereal, and no amount of any unfortified plant food.

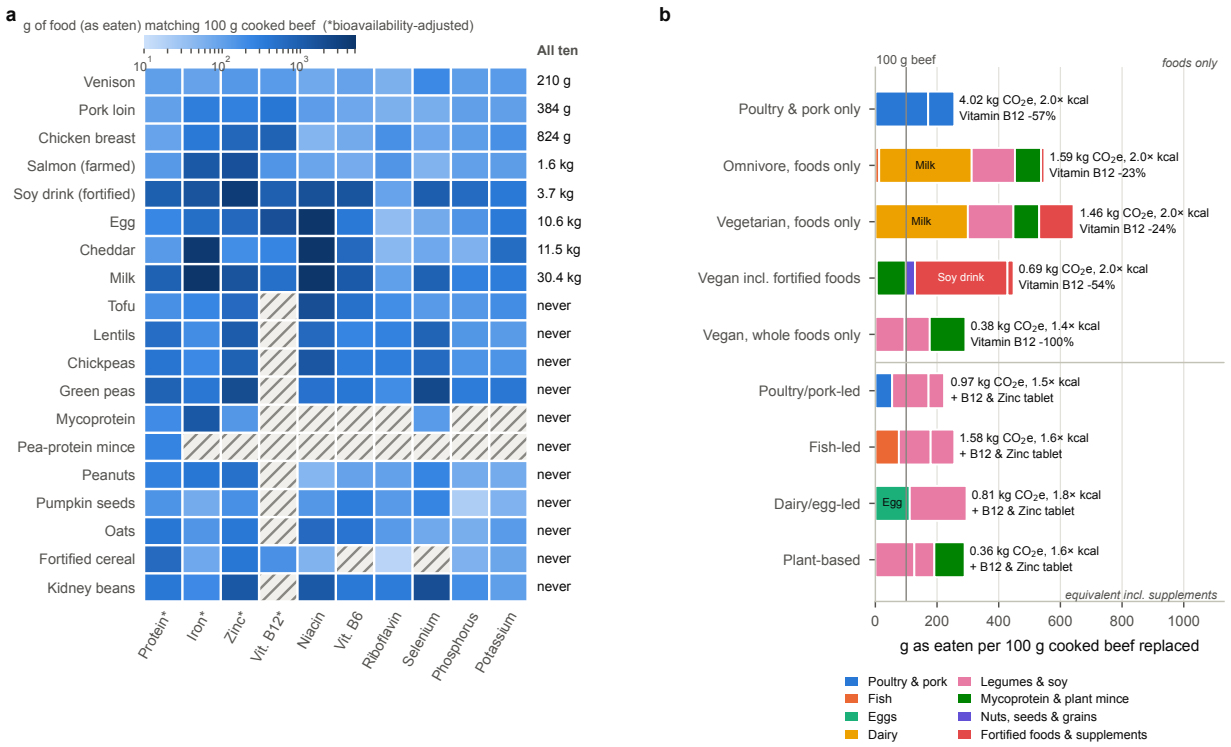


Fig. 1 | What it takes to replace the nutrients in 100 g of cooked beef. a, Grams of each food (as eaten) that supply the same amount of each nutrient as 100 g of cooked lean beef mince. Asterisks mark bioavailability-adjusted nutrients (DIAAS-weighted protein, absorbable iron, zinc and vitamin B_{12}). Hatched cells: the food contains none of the nutrient. The right-hand column gives the amount needed to match all ten nutrients at once. **b**, Minimum-emission baskets from linear programmes that match the key nutrients (quality-adjusted protein, absorbable iron, zinc and B_{12}, and selenium) within twice beef's energy, with saturated fat and sodium capped. The upper five use foods only; the lower four must be fully equivalent and may use B_{12} and zinc tablets. Labels give emissions per 100 g of beef replaced, energy relative to beef, supplements used and any remaining shortfall.

Of the nineteen alternatives, only wild venison (210 g) and pork loin (384 g) match all ten nutrients in less than half a kilogram. Imperial's own substitutes included venison and pork [18].

Optimising combinations of foods tells the same story (Fig. 1b, Table 1). Using foods alone and staying within twice beef's energy, no diet without ruminant meat matches all the key nutrients. Poultry and pork alone leave B_{12} vitamin B_{12} short by 57%. The best omnivore basket (salmon, milk, lentils, mycoprotein and fortified cereal) is still short by 23% and needs 2.3 times beef's energy to close the gap. Vegetarian and vegan food-only baskets cannot close it at any energy level given realistic portion caps. Adding a vitamin B_{12} tablet, and in some cases a small zinc supplement, changes this. All four nutritionally equivalent baskets, poultry/pork-led, fish-led, dairy/egg-led and plant-based, then match every key nutrient. They use 228–296 g of food and 1.5–1.8 times beef's energy. For example, the plant-based basket is 126 g kidney beans, 98 g mycoprotein, 65 g chickpeas, 0.3 b12 tablet, 0.1 zinc tablet. These baskets also deliver more fibre, potassium and phosphorus and less saturated fat than beef. The dairy/egg-led and plant-based baskets supply less niacin and vitamin B_6, partly because these are not reported for mycoprotein (Supplementary Table 5). Both are plentiful in UK diets. The answer to the biological question is therefore qualified. The same nutrition is obtainable, mainly from pulses, mycoprotein, fish, eggs, dairy and lean meats, but it takes about two to three times the food mass and 50–80% more energy. Matching beef's absorbable B_{12} and zinc in practice requires fortified foods or supplements. This is consistent with the lower zinc and B_{12} status observed in vegetarians and vegans [19,20].

Equivalent nutrition with a tenth of the emissions

We built the emission factor of beef eaten in the UK from its origin and production system (Methods; Extended Data Fig. 1). About 72% of supply is home-produced, roughly half of it from the dairy herd [21,22], and most imports come from Ireland. Farm footprints were taken from UK and FAO data [22–24] and converted to retail weight. The result is 34.7 kg CO_2e per kg of retail beef, close to Poore and Nemecek's average for UK producers (37.1 kg) and about half their global average of 66.4 kg for the same herd mix [2]. A 100 g portion of cooked beef therefore carries 5.59 kg CO_2e

Table 1 | Baskets replacing 100 g of cooked beef. Minimum-emission solutions with the key nutrients as targets and energy capped at 2.0 times beef's. "kcal ×" is the basket's energy relative to beef. "min. kcal ×" is the least energy at which the scenario can match all key nutrients ("none": impossible with realistic portions). Beef itself emits 5.59 kg CO_2e per 100 g (UK supply mix).

Scenario	Basket replacing 100 g cooked beef	kg CO_2e	kcal ×	min. kcal ×	Largest gap
Poultry & pork only	171 g chicken breast, 85 g pork loin	4.02	2.0	none	B_{12} −57%
Omnivore, foods only	300 g milk, 140 g lentils, 85 g mycoprotein, 12 g salmon, 11 g fortified cereal	1.59	2.0	2.3	B_{12} −23%
Vegetarian, foods only	300 g milk, 148 g lentils, 112 g soy drink, 84 g mycoprotein, 3 g fortified cereal	1.46	2.0	none	B_{12} −24%
Vegan incl. fortified foods	300 g soy drink, 94 g mycoprotein, 29 g pumpkin seeds, 20 g fortified cereal, 5 g lentils	0.69	2.0	none	B_{12} −54%
Vegan, whole foods only	116 g mycoprotein, 95 g kidney beans, 81 g lentils	0.38	1.4	none	B_{12} −100%
Poultry/pork-led	118 g chickpeas, 54 g chicken breast, 51 g lentils, 4 g mycoprotein, 0.4 zinc tablet, 0.3 b12 tablet	0.97	1.5	1.2	–
Fish-led	103 g lentils, 77 g kidney beans, 76 g salmon, 0.5 zinc tablet, 0.2 b12 tablet	1.58	1.6	1.6	–
Dairy/egg-led	184 g kidney beans, 112 g egg, 0.4 zinc tablet, 0.3 b12 tablet	0.81	1.8	1.6	–
Plant-based	126 g kidney beans, 98 g mycoprotein, 65 g chickpeas, 0.3 b12 tablet, 0.1 zinc tablet	0.36	1.6	1.3	–

(Monte Carlo 90% interval 5.8 (4.7–7.1)). The nutritionally equivalent baskets carry 0.36–1.58 kg, even though we deliberately used global mean footprints for every alternative food, which overstates UK-sourced pulses, poultry and dairy (Fig. 2a). Replacing beef with equivalent nutrition therefore cuts the associated emissions by 72–93%. The fish-led basket saves least (72%) and the plant-based basket most (93%).

Scaling up uses the national supply balance. The UK consumes 15.8 kg of beef per person a year in carcass-weight terms[21], which corresponds to 11.1 kg bought at retail and 6.9 kg eaten cooked (19 g a day). That beef emits 384 kg CO_2e per person a year. The equivalent baskets save 275–359 kg per person, about 16% of an average UK diet's footprint[25]. For London's 9.09 million residents[26], beef consumption emits 3.49 Mt CO_2e a year. Replacing it with equivalent nutrition would cut emissions by 2.5–3.3 Mt a year, or 2.8 (2.0–3.7) Mt for the poultry/pork-led basket including all parameter uncertainty (Fig. 2b). This is equivalent to 9–11% of London's 28.61 Mt territorial inventory for 2023[27] and 2.8–3.7% of its 88.0 Mt consumption-based footprint[28]. With global-average beef footprints the saving doubles to 5.7–6.4 Mt (20–23% of LEGGI). Counting only consequential emissions, the saving is 2.4 (1.7–3.2) Mt. Nationally the change is about 22 Mt a year, or 5.9% of UK territorial emissions. For comparison, enteric fermentation by UK beef cattle alone emits 11.3 Mt[29]. The largest source of uncertainty is how much beef people eat: survey-based intakes are about a third lower than the supply balance implies[30] (Extended Data Fig. 2).

How these numbers should be read matters. The "percentage of London" is a comparison of magnitudes, not a reduction in London's inventory. LEGGI counts emissions inside the city boundary, and London's livestock emits only about 0.01 Mt[27]. Almost all of the saving would occur on farms in the UK, Ireland and further afield. The consumption-based comparison is the like-for-like one: equivalent to a 3–4% cut in everything Londoners' consumption causes, or about a third of the 9.8 Mt attributed to their food[28]. The choice of climate metric also matters, because 55% of beef's footprint is methane. Under GWP_{20} the saving doubles (6.4 Mt). Under GWP*, which represents the effect of a sustained change in methane emission rate on warming[31,32], removing London's 68 kt of annual beef methane gives a cumulative warming-equivalent reduction of 186 Mt after 20 years. That is 3.3 times the GWP_{100} figure, and the two converge after a century (Fig. 2c). Whether GWP* is appropriate for attributing responsibility is debated[33,34], but it shows that ending a methane source buys near-term cooling. Finally, land freed from beef could store carbon. Searchinger and colleagues' carbon opportunity cost of 144 kg CO_2e per kg of beef[35] would add roughly 21 Mt a year for London's consumption. That is far larger than the production emissions, but realising it requires the freed land to be restored[36], so we do not count it in the headline numbers.

A campus ban is diluted by diners and markets

Taste Imperial bought 10.5 t of beef in 2022/23 and 8.5 t in 2023/24, and 0.07 t after removal (freedom-of-information records[37]; Supplementary Note 1). At 120 g per portion this is about 79,000 beef meals a year. A procurement inventory would book the removal of 293 t CO_2e a year, minus the replacement dishes: 232 t net. That is 0.06% of

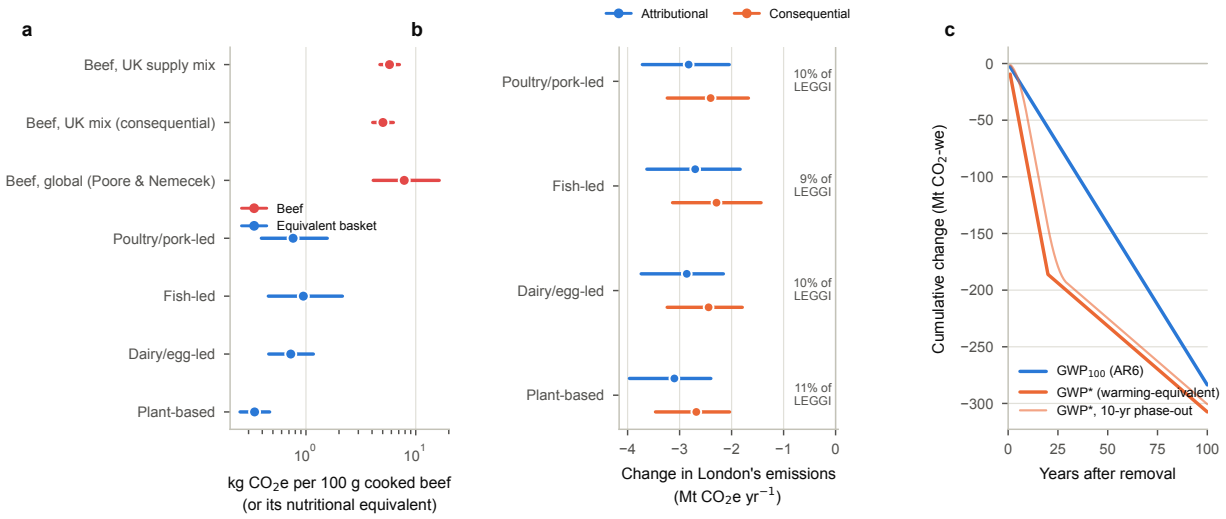


Fig. 2 | Emissions of beef and of its nutritionally equivalent replacements. **a**, kg CO_2e per 100 g of cooked beef, or per basket that replaces it. Dots are Monte Carlo medians and lines 90% intervals. "UK supply mix" combines six origin-by-system segments; "global" uses Poore and Nemecek's distributions for beef- and dairy-herd beef. **b**, Annual change in London's emissions if all beef eaten by Londoners were replaced by each basket. Attributional accounting counts beef's full footprint; consequential accounting counts only emissions that disappear when demand stops (dairy-origin beef keeps the cow's share). Text gives the attributional median as a share of London's 2023 territorial inventory (LEGGI). **c**, Cumulative change in warming-equivalent emissions after London stops eating beef, under GWP_{100} and under GWP*, which treats methane as a short-lived pollutant.

Imperial's 247 kt footprint[38] and about 7% of its catering emissions. Two responses outside the campus reduce this saving (Fig. 3).

First, some diners buy beef elsewhere. Evidence from canteens suggests this is small. Beef-free days at German universities did not change total or main-dish sales[39]. Doubling vegetarian availability in Cambridge colleges showed no rebound at the evening meal[40]. Removing meat options in Oxford left sales at eleven nearby cafeterias unchanged[41]. By contrast, whole meat-free days at a Swiss university cut on-campus sales by 17%[42]. We assume 7% of forgone beef meals are bought as beef off campus (Supplementary Table 15), which adds back 17 t.

Second, and more important, beef that Imperial no longer buys does not simply stay unproduced. In an integrated market, a fall in demand lowers the price slightly and other buyers absorb part of the freed supply. Only the share $\phi = \eta/(\eta - \varepsilon)$ of the demand cut reaches production, where η is the supply elasticity and ε the demand elasticity[43,44]. With UK beef demand elasticity of −0.59[45] and US-based supply elasticities of about 0.1–0.4 within a year and 1–5 within a decade[46,47], our multi-market equilibrium model gives a beef pass-through of 0.78 in the long run and 0.28 in the short run (Fig. 3a). The long-run value is close to Norwood and Lusk's widely used estimate of 0.68[44]. Market leakage therefore adds back 61 t in the long run. Replacement foods add 65 t, and their own market adjustment removes 6 t.

The net long-run effect is a reduction of 150 t CO_2e a year (90% interval 83–223 t). This is 52% of the beef's own footprint and 65% of what the inventory would report. In the first year the reduction is only 47 t (15–88 t). Because the replacement food's emissions are about a quarter of the beef's, the ban reduces emissions only if pass-through exceeds 0.25 (Extended Data Fig. 4). If beef production did not respond at all, the policy would increase emissions by about 65 t a year. At city scale the change, about 5.2 parts per million of London's territorial emissions, is too small to measure. Imperial's own records illustrate a further difficulty for evaluation. Chicken and pork purchases were already rising by about 12 t and 2 t a year before the ban, and they kept rising afterwards (48.5 t and 15.6 t in 2024/25). The emissions embodied in all meat purchases fell from 929 t to 814 t, but a before–after comparison would mix the policy with these trends (Supplementary Note 12).

What the ban costs diners, and what it does to their diets

With food prices held fixed, the only change for diners is a smaller choice set. In a random-utility model of dish choice, removing an option lowers each diner's expected utility by a log-sum amount[48,49]. We calibrate the model to a beef-dish own-price elasticity of about −1.5[50,51] and beef's 7.9% share of main meals. The loss per forgone beef meal is then £3.25 (£2.09–5.35) (Fig. 4a). Calibrated this way, the loss is almost exactly the meal price divided by the dish

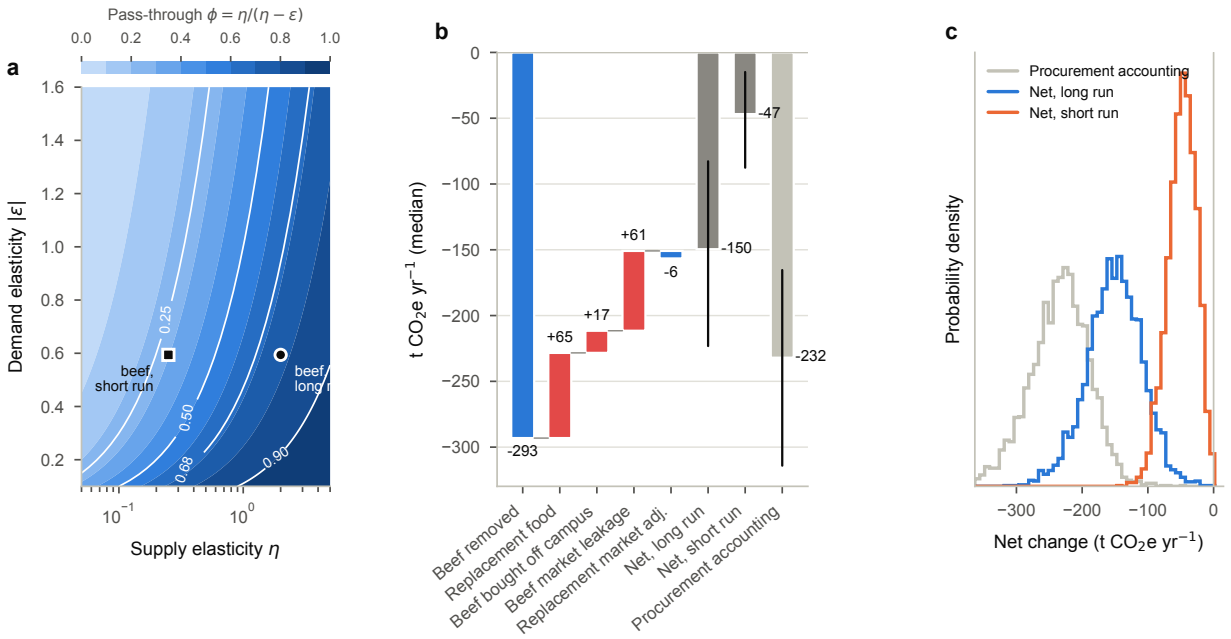


Fig. 3 | From campus purchases to atmospheric emissions. a, Share of a demand cut that reaches production, $\phi = \eta/(\eta - \varepsilon)$, as a function of supply (η) and demand (ε) elasticities. Markers show beef in the short and long run. **b**, Decomposition of the annual emissions change from removing beef at Imperial (long-run Monte Carlo medians). Grey bars show net outcomes, with 90% intervals, and what procurement accounting would report. **c**, Distributions of the net change (4,000 draws).

elasticity, whatever the strength of substitution within the meat nest (Supplementary Note 10). Because diners can always buy beef off campus, we also cap the loss at the cost of doing so, which gives £2.75 per meal. Across 79,000 beef meals the annual welfare loss is £214k (£145k–315k) (capped). The monetised climate benefit, at the UK's central carbon value of £277 per tonne[52], is £41k (£21k–68k) (Fig. 4b). The implied cost is £1,730 (£911–3,725) per tonne avoided (capped: £1,444). The welfare loss exceeds the climate benefit in essentially all Monte Carlo draws. The ban would break even only if diners valued a beef meal at no more than £0.52 over their next-best option.

This welfare estimate should be read as an upper bound on private cost rather than a verdict. It ignores habit formation, which erodes the loss over time. It ignores any value diners place on eating more sustainably. It also ignores health benefits of eating less red meat that diners do not fully take into account. All three can be measured (Supplementary Note 12). It is nonetheless consistent with other evidence that bans are a costly way to change diets: a one-off climate lecture reduced students' meat consumption for three years at about US$14 per tonne[53], and carbon labels[54] and more vegetarian options[40,55] reduce emissions while keeping choice.

The nutritional consequences for diners are small on average but not for everyone (Fig. 4c). Each forgone beef meal, replaced by Imperial's observed mix of chicken, pork, venison, fish, vegetarian and vegan dishes, provides 2.0 g more protein and 3.0 g more fibre. It provides 2.7 mg less zinc, 1.4 µg less vitamin B_{12} and 0.42 mg less iron. For a student who ate beef on campus 2 times a week, that means 11% less zinc and 27% less B_{12} relative to the reference intakes for women. These deficits are worth monitoring in groups whose iron and zinc intakes are already marginal[12]. Averaged over Imperial's 32,861 students and staff, the zinc deficit is only 0.02 mg a day.

Implications

Our results separate two claims that are often merged. The first is that replacing beef with other foods cuts emissions, and this is robust. The cut is 72–93% per portion when nutrition is matched, and it holds whichever basket is chosen and whether UK or global footprints are used. A city the size of London would avoid emissions equivalent to roughly a tenth of its territorial inventory, and even more under short-horizon metrics. The second claim is that a single institution achieves that cut by removing beef from its menus. This is much weaker. Market leakage, off-campus purchases and the emissions of substitute food cut the net saving to about half of the beef's footprint in the long run and about a sixth in the first year. A campus can report a large reduction in its procurement inventory while the effect on the atmosphere is smaller and, without a market response, could even be positive. This is the consequential-LCA point that a purchaser's inventory does not show the marginal effect of its decisions[56,57].

Three practical conclusions follow for Imperial and similar institutions. First, nutrition: removing beef should

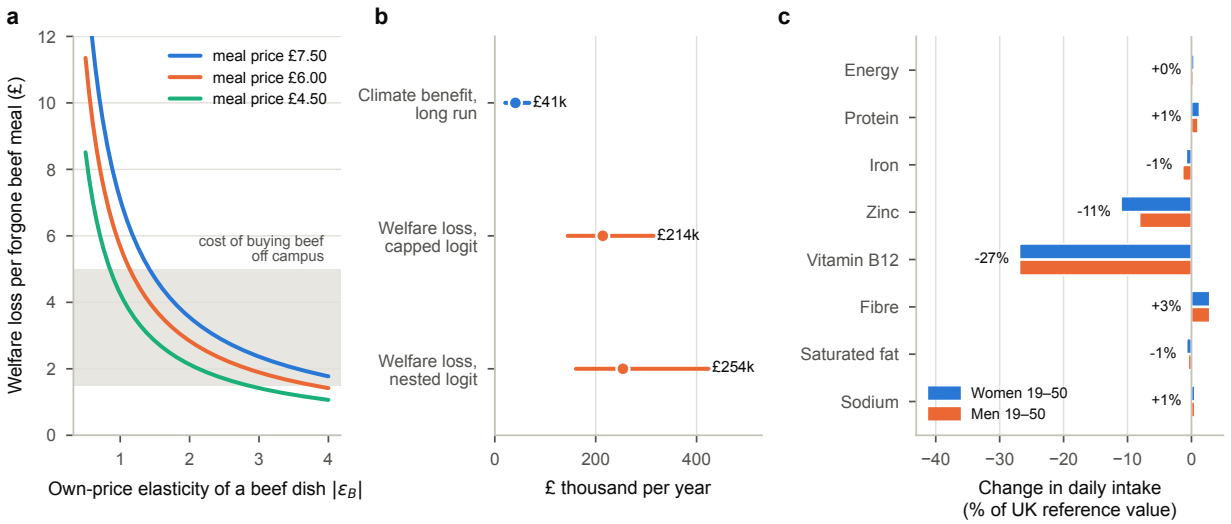


Fig. 4 | Consequences for diners. a, Welfare loss per forgone beef meal from a nested-logit model calibrated to the own-price elasticity of a beef dish, for three meal prices. The grey band is the cost of buying beef off campus instead, which caps the loss. **b**, Annual welfare loss with prices held fixed, compared with the monetised long-run climate benefit at the UK central carbon value (medians and 90% intervals). **c**, Change in average daily intake for someone who ate beef on campus 2 times a week, expressed as a share of UK reference values (RNI; fibre target; maximum for saturated fat and sodium).

come with appealing zinc- and B_{12}-rich alternatives. Mycoprotein, pulses with vitamin C, fish, eggs, dairy and fortified foods all qualify, and the mycoprotein and venison dishes Imperial already serves are good choices on these measures. Clear guidance on vitamin B_{12} is needed for students who stop eating meat. Second, accounting: claims should report consequential estimates alongside procurement inventories, with the short-run pass-through made explicit. Third, evaluation: the policy's effects on diets, satisfaction and welfare can be measured with till data, repeated 24-hour recalls[58], biomarkers and discrete-choice experiments[59]. Estimates should use designs that account for pre-existing trends[60–62] (Supplementary Note 12). Detecting a change in zinc intake of 0.5 mg a day needs about 770 participants per group, so studies must target frequent campus diners.

Our analysis has limitations. Nutrient equivalence is assessed per portion against one reference cut (lean mince). Bioavailability uses food-level absorption factors rather than meal-level models, and complementary proteins in mixed baskets are not credited. Emission factors for alternatives are global means, and market elasticities come from US and UK studies of whole national markets. The welfare model assumes the logit structure, and the dish elasticity that drives it has not been measured on any UK campus. Each assumption is exposed in the open code and data, and the Supplementary Information traces every number from its source to the result.

Methods

Overview and reproducibility

The model is written in Python (NumPy, SciPy, pandas). All inputs are in plain-text files in the repository's `data/` folder, each annotated with its source. One script reruns every calculation, writes all figures and tables, and generates the numbers quoted in this paper as LaTeX macros, so the text cannot drift from the code. Supplementary Information Notes 1–13 give every equation, parameter and intermediate result. Uncertainty is propagated by Monte Carlo simulation (20,000 draws for London, 4,000 for the campus); we report medians with 90% intervals.

Food composition and bioavailability

Nutrient contents per 100 g as eaten were taken from USDA FoodData Central SR Legacy[63] for 16 foods (FDC identifiers in Supplementary Table 3), from manufacturer labels and analyses for mycoprotein[15,16] and plant-based mince, and from UK labels for fortified cereal and soya drink. The reference is 90%-lean beef mince, pan-browned. Protein is weighted by truncated DIAAS from pig and rat assays[11,13,64–68]. Iron is split into haem and non-haem pools absorbed at 25% and 10%[69,70]. Zinc absorption is 35% from animal foods, 25% from moderate-phytate foods and 18% from pulses, seeds and whole grains[14,71,72]. Vitamin B_{12} absorption is 60% from meat, 42% from fish, 65% from dairy, 9% from eggs and 50% from fortificants[17]. Ranges for all factors are tested in sensitivity analyses. UK reference nutrient intakes follow COMA[73].

Single-food equivalents and diet optimisation

For food i and nutrient k, the mass matching 100 g of beef is $100\,b_k/a_{ik}$, where b_k is beef's content and a_{ik} the food's. Baskets are found by linear programming (HiGHS via SciPy). The programme minimises emissions subject to: supply of each key nutrient at least equal to beef's; energy between 0.85 and 2.0 times beef's; saturated fat no higher than beef's; sodium no more than 150 mg above beef's; and portion caps. Each nutrient constraint has a penalised slack variable, so unavoidable shortfalls are reported rather than making the problem infeasible. Scenarios restrict the allowed food groups. "Equivalent" scenarios also require at least 60% (dairy/egg-led: 50%) of quality-adjusted protein to come from the named group and allow B_{12} and zinc tablets. A minimum-energy variant gives the least energy at which each scenario can match beef. Sensitivity analyses cover bioavailability (plant-favourable and plant-unfavourable extremes), no bioavailability adjustment, a strict ±15% energy band, and matching all ten nutrients (Extended Data Fig. 3).

Emission factors

For alternative foods we use Poore and Nemecek's global mean emissions per kg of retail product as central values and their 5th, 50th and 95th percentiles to fit log-normal distributions[2]. For foods they do not cover we use verified product footprints: mycoprotein[74] and plant-based mince[75,76]. Cooking yields come from USDA and Bognár tables[77,78], and a household waste fraction of 10% applies to all foods[79]. For beef we combine six supply segments: UK suckler, UK dairy-origin, Ireland, other Europe, Oceania and South America (Supplementary Table 8). Their shares come from Defra supply statistics and import data[21]. Farm-gate footprints per kg of carcass come from Defra/CIEL, PRISM (Agrecalc) and FAO GLEAM data[22–24]. These are converted to retail weight with a yield of 0.70 and a post-farm-gate adder of 3.0 kg CO_2e per kg. In the consequential variant, dairy-origin beef carries only the rearing and finishing share of its footprint (avoidable fraction 0.7), because the cow's emissions are reallocated to milk rather than avoided[56].

Scaling, metrics and Monte Carlo

Per-capita beef supply (15.8 kg carcass weight a year; production plus imports minus exports divided by population[21,26]) is converted to retail purchases and cooked intake as above. Annual emission changes are $10\,B_{\text{eaten}}\text{GHG}_b - B_{\text{retail}}\text{EF}_{\text{beef}}$ per person and are scaled by London's and the UK's populations. They are compared with LEGGI 2023 (territorial)[27], London's consumption-based footprint for 2022[28] and UK national totals[29]. Metric conversions use each footprint's gas composition and the GWP vintage in which it was reported, converted to AR6 values[80]. Warming-equivalent emissions use GWP* with the coefficients of Smith et al.[32]. The Monte Carlo samples consumption, yields, waste, segment footprints, segment shares (Dirichlet), avoidable fractions and every alternative-food footprint jointly. One-at-a-time sensitivities are shown in Extended Data Fig. 2.

Market model

The campus shock is propagated through an equilibrium displacement model of nine UK food markets (beef, lamb, pork, poultry, fish, eggs, dairy, plant protein and venison)[43,81]. Each market has constant-elasticity supply and demand with own- and cross-price elasticities. A demand shift ΔD at constant prices changes prices by $(\text{diag}(\eta) - \mathcal{E})^{-1}\Delta D/Q$ and production by $Q\eta\,\mathrm{d}\ln P$ (Supplementary Note 9). Demand elasticities come from UK estimates and meta-analyses[45,82,83], including UK cross-price elasticities among meats[45]. Short-run (1-year) and long-run (10-year) supply elasticities come from US estimates[46,47], moderated for an integrated European market, and are sampled from triangular distributions (Supplementary Table 11). The shock is built from Imperial's beef purchases (9.5 t a year, freedom-of-information data[37]), the share of forgone beef meals going to each alternative dish type or off campus (Dirichlet-distributed; Supplementary Table 15), and dish recipes. Beef uses the consequential emission factor (31.0 kg CO_2e per kg).

Welfare and nutrient intake of diners

Diners choose dishes in a nested logit with beef and other meat dishes in one nest (dissimilarity λ). With prices fixed, the compensating variation per meal occasion is $\ln\left(1 - P_M + P_M(1 - P_B/P_M)^{\lambda}\right)/\alpha$[48,49]. The marginal utility of money α is calibrated so that the beef dish's own-price elasticity equals ε_B. The loss per forgone beef meal is capped at the cost of buying beef off campus. Losses are compared with the avoided emissions valued at DESNZ carbon values (£139–416 per tonne, central £277; 2022 prices)[52]. Nutrient changes per forgone beef meal use the same dish recipes and composition data.

Use of AI tools

The analysis code, data assembly and manuscript drafting were carried out with the assistance of an AI system (Claude, Anthropic). Every numerical input is traceable to the sources cited in the data files, and the author is responsible for the content.

Data availability

All input data, with sources, and all generated results are in the accompanying repository (`data/`, `results/`). The FOI purchase records are published by Imperial College London[37].

Code availability

The code that reproduces every figure, table and number is in the repository (`src/beefwhatif`, `scripts/run_all.py`) under the MIT licence.

Competing interests

The author declares no competing interests.

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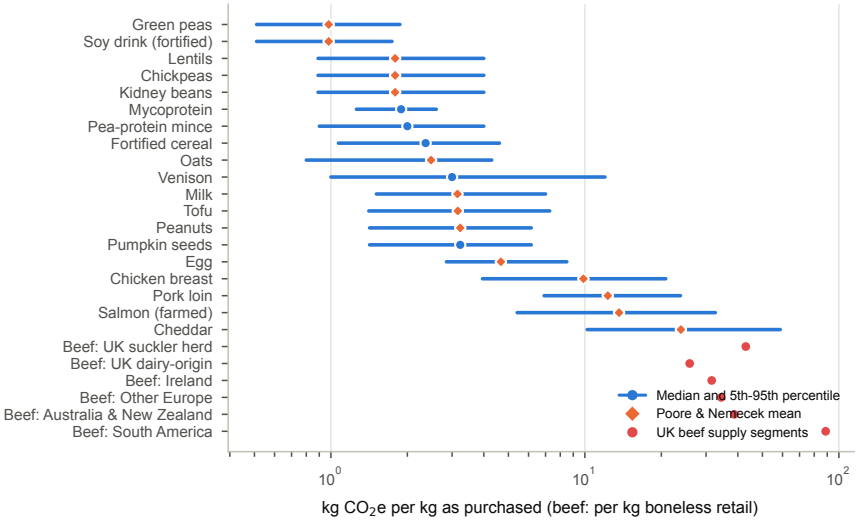
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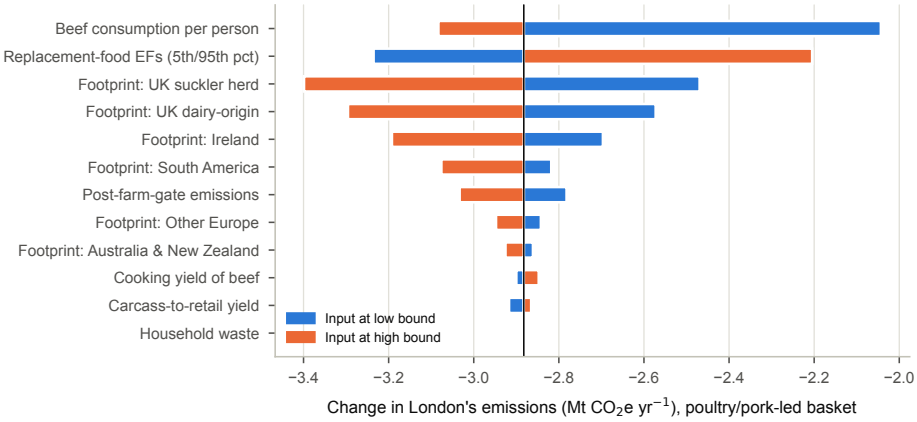
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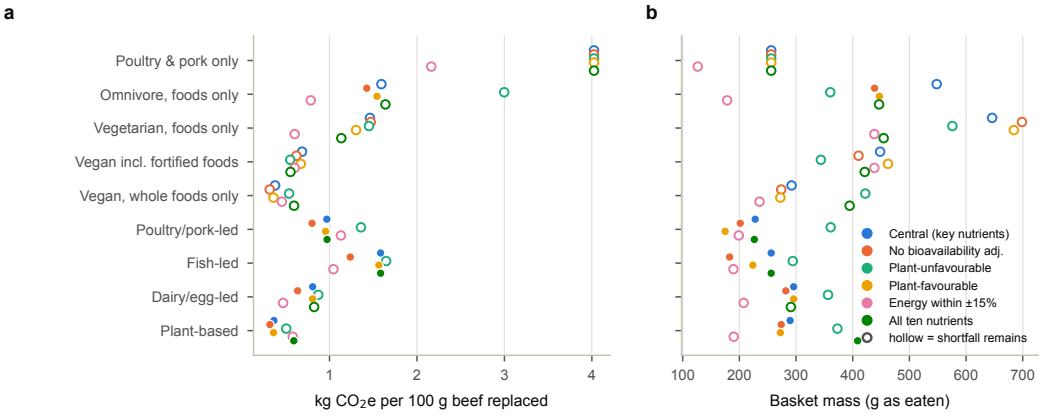
Extended Data



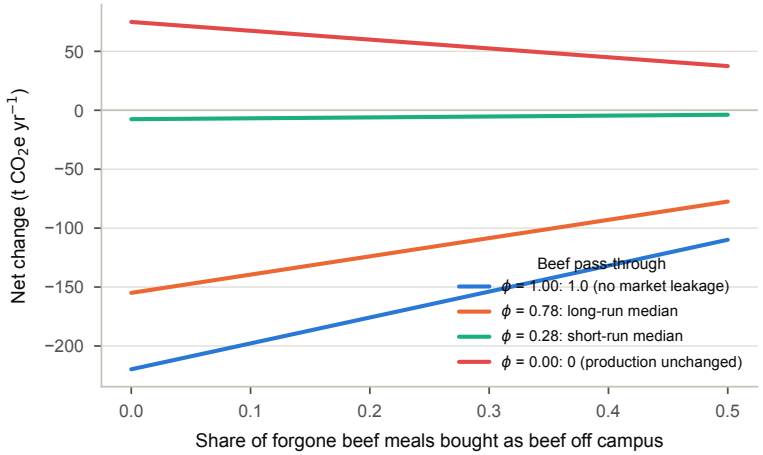
Extended Data Fig. 1 | Emission factors. kg CO_2e per kg of each food as purchased: median and 5th–95th percentile range used in the Monte Carlo, with Poore and Nemecek's global means. Red dots: retail emission factors of the six UK beef supply segments; their consumption-weighted mean is 34.7 kg CO_2e per kg.



Extended Data Fig. 2 | Sensitivity of London's emissions change. Each input is moved to its low and high bound with all others at central values (poultry/pork-led basket, UK supply mix). Beef consumption per person dominates. Its low bound reproduces survey-based intakes.



Extended Data Fig. 3 | Robustness of the optimised baskets. a, Emissions and **b**, mass of each scenario's basket under alternative bioavailability assumptions, a strict isocaloric constraint and matching all ten nutrients. Hollow markers: a shortfall remains.



Extended Data Fig. 4 | Campus net change, beef pass-through and off-campus purchasing. Net annual change for Imperial as the share of forgone beef meals bought as beef off campus varies, for pass-through values from 0 to 1. The sign depends on pass-through alone: the ban reduces emissions only if pass-through exceeds the ratio of replacement-food to beef emissions (about 0.25).

Extended Data Table 1 | Decomposition of the campus emissions change (t CO_2e yr^{-1}). Monte Carlo median and 90% interval.

Component	Long run	Short run	Long run, no cross-price
Beef removed from menus	−293 (−373 to −233)	−294 (−374 to −233)	−293 (−373 to −236)
Replacement food bought	65 (42 to 103)	64 (41 to 103)	64 (43 to 104)
Beef bought off campus	17 (3 to 51)	18 (3 to 51)	18 (3 to 51)
Beef market leakage	61 (35 to 109)	197 (148 to 261)	61 (36 to 111)
Replacement market adjustment	−6 (−14 to −0)	−36 (−63 to −21)	−12 (−20 to −7)
Net change	−150 (−223 to −83)	−47 (−88 to −15)	−154 (−224 to −94)
Procurement accounting	−232 (−314 to −165)	−233 (−316 to −166)	−232 (−312 to −168)
Beef pass-through ϕ	0.78 (0.62 to 0.87)	0.28 (0.18 to 0.38)	0.78 (0.62 to 0.86)

Extended Data Table 2 | Welfare and monetised climate effects for Imperial diners.

Quantity	Median	90% interval
Loss per forgone beef meal, nested logit (£)	3.25	2.09 to 5.35
Loss per forgone beef meal, capped (£)	2.75	1.92 to 3.92
Annual welfare loss, nested logit (£)	253,969	160,709 to 423,618
Annual welfare loss, capped (£)	214,059	144,507 to 314,739
Emissions avoided, long run (t CO_2e)	150	83 to 223
Monetised climate benefit (£)	40,818	20,608 to 68,436
Welfare cost per tonne avoided (£/t)	1,730	911 to 3,725
Welfare cost per tonne, capped (£/t)	1,444	818 to 2,896
P(capped loss > climate benefit)	100%	